

QA Test Guide — Step by Step

Printable walkthrough generated from [docs/QA_TEST_PLAN.md](#). Each case shows its Gherkin scenario and the plain-English steps. Record results on the companion [QA Run Log](#) sheet. Generated 2026-07-02.

End-to-end manual test plan covering the **entire** functionality of the app across all three clients: **Web** (Blazor WASM), **Desktop** (MAUI / Windows), and **Android** (MAUI). Each case is written twice — a **Gherkin** scenario (Given/When/Then, matching this project's convention so it can later seed the automated E2E . Tests Playwright suite) and a ****plain-English walkthrough**** a manual tester can follow step by step.

Brand in examples is "Perezosoft"; substitute your app's brand if rebranded.

How to use this document

- **Run the Smoke suite (§4) first.** It's the ~15-minute critical path. If any smoke case fails, stop and report — deeper suites will likely cascade.
- **Each case has an ID** (e.g. QA-AUTH-03). Record the result against the ID using the **sign-off sheet (§14)**: Pass / Fail / Blocked / N-A, plus tester, build/commit, date, notes.
- **Priority:** ● **Smoke** Smoke (critical path) · ● **Core** Core (run every regression) · ● **Edge** Edge (run on full regression or when the area changed).
- **Both formats describe the same test.** Read whichever suits you; the Gherkin is the source of truth for automation.
- **"App" = whichever client the suite header names.** Most behavior is identical across clients (same API, same shared RCL UI); the per-client suites (§11–12) only cover what genuinely differs — the auth transport and session persistence.

1. Environment & prerequisites

1.1 Backing services + API (host machine)

```
docker compose up -d # Postgres + Mailpit
dotnet run --project src/Api --launch-profile https # binds https:7160 (web/desktop) AND http:5238 (android)
```

- API health/liveness check: `curl -k https://localhost:7160/health → 200` (Healthy);
`curl -k https://localhost:7160/health/ready → 200` when the database is reachable (503 if not). *(The older `curl -k -X POST https://localhost:7160/api/auth/refresh → 401` reachability check still works.)*
- **Mailpit UI:** <http://localhost:8025> — this is the dev mail trap. Every magic link, OTP code, and invitation email lands here. Keep it open in a tab throughout testing.

■ Email only reaches Mailpit if SMTP points at it. If your repo-root .env has the Email__Smtp__* lines set to a real provider (e.g. Brevo), the API sends auth emails there and Mailpit stays empty — every email-based case below will appear to "fail." For QA, route mail to Mailpit by either:

- commenting out the Email__Smtp__* lines in .env (unset → defaults to Mailpit localhost:1025), or
- leaving .env untouched and overriding on the command line (command-line config beats .env):

```
```bash
dotnet run --project src/Api --launch-profile https -- \
--Email:Smtp:Host=localhost --Email:Smtp:Port=1025 --Email:Smtp:Username= --Email:Smtp:Password=
```
```

Verify by triggering one OTP (QA-SMK-01) and confirming it appears in Mailpit before running the suite.

■ Email delivery is now asynchronous. Requesting a code/link/invite enqueues the email and a background dispatcher (the outbox) sends it — so it appears in Mailpit **a few seconds later, not instantly. Wait briefly before assuming failure. The send request now always returns success** (reliability moved to the background): if an email never arrives while SMTP points at Mailpit, the message is retrying or dead-lettered in the OutboxMessages table — it is no longer surfaced as a request error.

- Web app: dotnet run --project src/Web --launch-profile https → <https://localhost:7008>.

■ Always use the https profile for both web and API. Chrome treats http://localhost and https://localhost as different sites, so the refresh cookie is dropped over http and sign-in silently fails to persist. (See docs/DECISIONS.md / the schemeful-same-site note.)

■ The passwordless endpoints are rate-limited (default 5 requests/minute per IP on /otp/send, /magic-link/send, and /otp/verify). If you fire many code/link requests or verify attempts in quick succession you may get HTTP 429 — that's the abuse guard working (QA-AUTH-11), not a bug. Pace requests, or wait ~1 minute for the window to reset.

1.2 Test accounts & data

| Need | What to prepare |
|--------------------------------|--|
| Two real Google accounts | e.g. qa.owner@gmail.com, qa.member@gmail.com — for OAuth + linking + multi-user household flows. |
| One Microsoft personal account | for the Microsoft OAuth path (provider pinned to the *consumers* tenant). |
| Throwaway email addresses | any address works for magic-link / OTP — mail is trapped by Mailpit, so the address need not be real. Use distinct ones per test to keep inboxes clean. |
| Two browser contexts | a normal window and an incognito/second-profile window. The household invite flow needs two *different* signed-in users at once; incognito gives you an isolated session + cookie jar. |

Database reset between full runs (optional but recommended): to retest "new user" onboarding cleanly, you need users that don't yet exist. Either use fresh email addresses each run, or reset the dev DB: docker compose down -v && docker compose up -d, then re-apply migrations by starting the API. A volume wipe destroys all test data — only do it on the dev environment.

1.3 Desktop (MAUI Windows) — additional setup

- Run the app from Visual Studio (Windows Machine target) or dotnet build src/Maui -t:Run -f net10.0-windows....
- API must be running on https://localhost:7160 (the desktop client's base URL).
- OAuth uses a loopback browser flow — your default system browser will open a tab during OAuth.

1.4 Android (MAUI) — additional setup

Follow docs/MOBILE_TESTING.md. The essential bits:

- Emulator (AVD) or USB device running.
- **adb reverse tcp:5238 tcp:5238 — re-run every time the device/emulator restarts** (it does not persist). Verify with adb reverse --list.

- API started with the **https** profile (binds the cleartext : 5238 leg the device uses).
- Provider redirect URIs registered: <http://localhost:5238/signin-google> and <http://localhost:5238/signin-microsoft>.

2. Scope

In scope: authentication (all methods, all clients), new-user onboarding, household/tenant management, invitations & joining, account settings (linked providers), localization (EN/ES), transactional emails, and cross-cutting security (tenant isolation, auth guards, token lifecycle).

****Out of scope (per docs/PROJECT_BRIEF.md OUT list & current state):**** SMS OTP, OAuth providers beyond Google/Microsoft, FR/DE/PT languages (scaffolded but not translated — see docs/LOCALIZATION.md), and any app-specific domain features not yet built on this template.

Platform services with no client UI (API-/operational-level, not manually testable through the app yet): the billing API (`/api/billing/*`), the append-only audit log, OpenTelemetry telemetry, the health endpoints, the background outbox/inbox/scheduled-jobs, and **file storage** — the `IFileStorage` seam + the signed download endpoint `GET /api/files/{token}` (anonymous, the token `*is*` the authorization; local-disk only — cloud backends hand out native presigned URLs). These are **covered by automated tests** (`tests/Api.Tests`); E2E is pending. Health has a smoke check (QA-SMK-07); manual cases for the rest will be added when client UI exists. **GDPR data export** (`POST /api/household/export`) and **account erasure** (`DELETE /api/auth/me`) now **have a web UI** (UI-1: owner Household → Data, Settings → Danger zone) — covered by the manual cases QA-HH-13 + QA-SET-07. **RBAC role management now has a web UI** (RBAC-3) — covered by the household cases QA-HH-09..12. **MFA now has a web UI** (UI-2) — enrollment/ disable in Settings and the sign-in step-up on Login — covered by QA-MFA-01..03. The **in-app notification center now has a web UI** (UI-3) — the header bell (list, unread count, mark-read) and Settings delivery-preference switches — covered by QA-NOTIF-01..03. The **platform-staff admin surface now has a web UI** (UI-4) — a staff-only `/admin` console (tenant list/detail + impersonation) — covered by QA-ADMIN-01..03. The **public API (PUBAPI)** and **outbound webhooks (HOOKS)** are intentionally UI-less (they're for machines) and **config-gated off** — they have **manual curl/Postman cases in §14b** (QA-API-01..06), in addition to automated tests.

3. Test data conventions

- **Owner user** = the account you sign in with first; it auto-creates and owns a household.
- **Member user** = a second account invited into the owner's household.
- Household auto-named on creation; rename it to something recognizable (e.g. "QA House") early so you can spot it in the header tenant badge.

4. Smoke suite ● Smoke (run first — ~15 min)

The critical path: can a user get in by each method, reach the app, and get out.

QA-SMK-01 — Web: OTP sign-in happy path ● Smoke (Web)

Gherkin

```
Given I am an anonymous user on the /login page of the web app
When I enter my email and request a 6-digit code
And I retrieve the code from Mailpit and submit it
Then I am signed in and land on the home page
And the header shows my household name and my display name
```

Walkthrough

- 1 Open <https://localhost:7008> → you're redirected to `/login`.
- 2 In the email field enter `qa-smoke@example.com`; click **Email me a 6-digit code**.
- 3 **Expected:** the form switches to a code-entry view ("Enter the code we sent to...").
- 4 Open Mailpit (<http://localhost:8025>); open the newest mail; copy the 6-digit code.
- 5 Enter the code; click **Verify code**.

- 6 **Expected:** you land on the home page; the top header shows a tenant badge (household name) and your display name, plus **Household**, **Settings**, **Sign out** buttons.

QA-SMK-02 — Web: Google OAuth sign-in ● Smoke (Web)

Gherkin

```
Given I am on the /login page
When I click "Continue with Google" and complete Google consent
Then I am returned to the app, signed in, on the home page
```

Walkthrough

- 1 From /login, click **Continue with Google**.
- 2 **Expected:** full-page redirect to Google's consent screen.
- 3 Choose qa.owner@gmail.com, approve.
- 4 **Expected:** redirected back into the app, signed in, on the home page with the header chrome.

QA-SMK-03 — Web: Sign out ● Smoke (Web)

Gherkin

```
Given I am signed in to the web app
When I click "Sign out"
Then I am returned to the /login page
And navigating to /settings redirects me back to /login
```

Walkthrough

- 1 While signed in, click **Sign out** in the header.
- 2 **Expected:** you land on /login.
- 3 In the address bar go to /settings.
- 4 **Expected:** you're bounced back to /login (no access without a session).

QA-SMK-04 — Web: Session persists across reload ("remember me") ● Smoke (Web)

Gherkin

```
Given I am signed in to the web app
When I reload the page (or reopen the tab)
Then I am still signed in without re-authenticating
```

Walkthrough

- 1 Signed in, press F5 / reload.
- 2 **Expected:** brief load, then the app shell — still signed in, no trip to /login. (A silent refresh exchanges the refresh cookie for a new access token on load.)

QA-SMK-05 — Desktop: OTP sign-in ● Smoke (Desktop) — see QA-DSK-01

QA-SMK-06 — Android: OTP sign-in ● Smoke (Android) — see QA-AND-01

QA-SMK-07 — API health & readiness ● Smoke (Platform)

Gherkin

```
Given the API is running
When I GET /health and /health/ready
Then /health returns 200 "Healthy"
And /health/ready returns 200 when the database is reachable, 503 when it is not
```

Walkthrough

- 1 `curl -k https://localhost:7160/health → 200`, body Healthy (liveness — process up).
- 2 `curl -k https://localhost:7160/health/ready → 200` (readiness — Postgres reachable).
- 3 **(Optional)** stop the DB (`docker compose stop db`) and re-run step 2 → **503**; then `docker compose start db` and confirm it returns to **200**.

- 1 **Expected:** liveness is 200 whenever the process runs; readiness tracks DB reachability. Responses are **status-only** (no connection details leaked).

5. Web — Authentication ● Core

QA-AUTH-01 — Magic-link sign-in happy path ● Core (Web)

Gherkin

```
Given I am an anonymous user on /login
When I enter my email and request a magic link
And I open the emailed link from Mailpit
Then I am signed in and landed in the app
```

Walkthrough

- 1 On /login enter qa-magic@example.com; click **Send me a magic link**.
- 2 **Expected:** a success panel — "Check your inbox... we sent a link to qa-magic@example.com" — with a **Use a different email** link.
- 1 In Mailpit open the newest mail; click the sign-in link (or copy it into the same browser).
- 2 **Expected:** the link resolves and you end up signed in, in the app.

QA-AUTH-02 — Microsoft OAuth sign-in ● Core (Web)

Gherkin

```
Given I am on /login
When I click "Continue with Microsoft" and complete consent
Then I am returned signed in
```

Account resolution (read before testing): signing in with a provider whose email matches an existing account links the provider to that account rather than creating a duplicate — but only when the email is **verified**. Google asserts this; Microsoft on the ****consumers**** (personal MSA) tenant is trusted to have a verified email even though it omits the claim (IProviderEmailTrust). For **work/school** tenants (organizations/common/a tenant GUID) and any other provider, a same-email auto-link is **refused** (fail-closed takeover guard, audit MITI-3) — the user signs in with their original method and links the provider from **Settings** instead (QA-SET-02). To test the plain first-time path below, use a Microsoft account whose email has **no** prior account.

Walkthrough

- 1 Click **Continue with Microsoft**; sign in with a **personal** Microsoft account.
- 2 **Expected:** returned to the app signed in. A brand-new email creates a new account; a personal Microsoft account whose email already exists auto-links to that account (consumers tenant). (If you see a tenant/reply-URL error, the provider registration is the cause — out of app scope, note it.)

QA-AUTH-03 — OTP wrong code is rejected ● Core (Web)

Gherkin

```
Given I requested an OTP code for my email
When I submit an incorrect 6-digit code
Then I see an "incorrect code" error and remain on the code-entry screen
```

Walkthrough

- 1 Request an OTP (as QA-SMK-01) but **do not** use the real code.
- 2 Enter 000000; click **Verify code**.
- 3 **Expected:** inline error ("code is incorrect / verification failed"); you stay on the code-entry view and can retry. You are **not** signed in.

QA-AUTH-04 — OTP cumulative lockout & code expiry ● Edge (Web)

Gherkin

Given I requested an OTP code
When I submit wrong codes up to the cumulative limit (default 5 failures per email within a 15-min window)
Then the email is locked – further attempts, INCLUDING a freshly requested code, are rejected until the window elapses
And separately, an unused code is rejected once it passes its lifespan (default 10 minutes)

Walkthrough

- 1 Request an OTP for an email; enter a **wrong** 6-digit code until you hit the cumulative limit (default 5, `Auth:Otp:MaxAttempts`, counted ****per email across the `Auth:Otp:LockoutWindowMinutes` = 15-min window****, not per code).
- 1 **Expected:** after the limit, a "too many attempts" lockout error.
- 2 Now request a **fresh** code and submit it — even the **correct** one.
- 3 **Expected:** still rejected — requesting a new code does **NOT** reset the budget (this is the brute-force defense; a resend can't hand the attacker another N guesses). The lock clears after the 15-min window.
- 1 **Expiry sub-case:** separately, request a code and wait past `Auth:Otp:CodeLifespanMinutes` (default 10) — the stale code is rejected.

Note (enumeration): OTP verify returns the **same** generic "invalid code" error whether the code was wrong or there is no active code — it never reveals whether an address has an outstanding OTP. *(Long-wait cases — lower the config to test fast. Mind the per-IP rate limit, QA-AUTH-11: 5 wrong verifies in a minute also trips the 429 throttle.)*

QA-AUTH-11 — Passwordless endpoints are rate-limited ● Core (Web)

Gherkin

Given I rapidly request codes/links (or verify attempts) for the same client
When I exceed the per-IP limit (default 5 per minute)
Then further requests are rejected with HTTP 429 until the window resets

Walkthrough

- 1 From `/login`, request an OTP (or magic link) repeatedly in quick succession — more than 5 within a minute.
- 1 **Expected:** after the limit the request is throttled (**HTTP 429**, surfaced as a "try again shortly" / send-failed state). The same throttle applies to **OTP verify**.
- 1 Wait ~1 minute; requests succeed again.

Protects against email-bombing and OTP brute-forcing. Expected behavior, not a defect — see the §1.1 pacing note.

QA-AUTH-05 — Magic link is single-use ● Edge (Web)

Gherkin

Given I signed in by clicking a magic link
When I click the same link a second time
Then it is no longer valid

Walkthrough

- 1 Complete QA-AUTH-01. Then revisit the **same** link from Mailpit.
- 2 **Expected:** it does not grant a second session — you're sent to `/login` with an invalid-link indication (or simply not signed in). The token is consumed on first use.

QA-AUTH-06 — Magic link expiry ● Edge (Web)

Walkthrough: request a link, wait past `Auth:MagicLink:TokenLifespanMinutes` (default 15), then open it.
Expected: rejected as expired; user lands on `/login`. *(Lower the config to test fast.)*

QA-AUTH-07 — User-enumeration protection ● Core (Web)

Gherkin

```
Given an email address that has no account
When I request a magic link or OTP for it
Then the UI shows the same success/"check your inbox" response as for a real account
```

Walkthrough

- 1 Request a magic link **and** an OTP for a never-before-used address.
- 2 **Expected:** identical success messaging to a known address — the app never reveals whether an account exists. (Mailpit will still show whatever the system chose to send.)

QA-AUTH-08 — Login error banners ● Edge (Web)

Gherkin

```
Given the login page is loaded with an error query parameter
Then a human-readable error banner is shown
```

Walkthrough — visit each URL and confirm a red banner with sensible copy:

- `/login?error=external_failed` → external sign-in failed message.
- `/login?error=email_unverified` → email-unverified message.
- `/login?error=invalid_link` → invalid/expired link message.
- `/login?error=somethingelse` → generic "something went wrong" fallback.

QA-AUTH-09 — Email-format validation ● Edge (Web)

Walkthrough: on `/login`, click a send button with an empty or malformed email (e.g. `abc`). **Expected:** inline "enter a valid email" validation; no request sent.

QA-AUTH-10 — Magic link & OTP available on web; OAuth always ● Edge (Web)

Walkthrough: confirm the web login page shows **both** "Send magic link" and "Email me a code", plus Google/Microsoft buttons. (Native clients hide magic link — covered in §11–12.)

6. Web — Onboarding & new tenant ● Core

QA-ONB-01 — First-ever sign-in auto-creates a household, user is owner ● Smoke (Web)

Gherkin

```
Given an email/account that has never signed in before
When I authenticate for the first time (any method)
Then a new household is created and I am its owner
And the header shows that household and my name
```

Walkthrough

- 1 Sign in with a brand-new account/email (fresh address or post-DB-reset).
- 2 **Expected:** you reach the app immediately (no "create household" prompt — provisioning is automatic). Open **Household:** you are listed with the **Owner** badge and are the only member.

QA-ONB-02 — Returning user keeps their household ● Core (Web)

Walkthrough: sign out and back in with the same account. **Expected:** same household, same role, data intact.

7. Web — Household management ● Core

Roles: **owner** > **admin** > **member** (ADR-009). **Owner + admin** can rename and invite/remove members; **owner-only**: change member roles (promote/demote), transfer ownership, dissolve. Members see a read-only name and a **Leave** button. The owner's row never shows action buttons.

QA-HH-01 — Owner renames the household ● Core (Web)

Gherkin

```
Given I am the household owner on /household
When I change the name and click Rename
Then the new name is saved and reflected in the header tenant badge
```

Walkthrough

- 1 **Household** → edit the name field (e.g. "QA House") → **Rename**.
- 2 **Expected:** success banner; the header tenant badge updates to the new name (a silent refresh updates the claim).

QA-HH-02 — Member sees read-only household, no owner controls ● Core (Web)

Precondition: signed in as a *member* (use QA-INV flow to create one). **Walkthrough:** open **Household**. **Expected:** household name shown as read-only heading; no rename/invite/transfer controls; a **Leave** button is present.

QA-HH-03 — Owner removes a member ● Core (Web)

Gherkin

```
Given I am the owner and another member exists
When I click Remove next to that member and confirm
Then they are removed from the household member list
```

Walkthrough

- 1 As owner with a member present, click **Remove** by the member's row.
- 2 **Expected:** a confirm dialog ("Remove <name>?"); on confirm, success banner and the member disappears from the list. (The removed user is re-homed to a fresh solo household — verify by signing in as them: they now own an empty household — see QA-HH-08.)

QA-HH-04 — Owner cannot remove themselves ● Edge (Web)

Walkthrough: as owner, confirm there is **no Remove button on your own row**. (Owner departs via transfer or dissolve, not removal.)

QA-HH-05 — Transfer ownership ● Core (Web)

Gherkin

```
Given I am the owner and at least one other member exists
When I select that member and click Transfer
Then they become owner and I become a regular member
```

Walkthrough

- 1 As owner with ≥ 1 other member, in **Transfer ownership** pick the member → **Transfer**.
- 2 **Expected:** success banner. The chosen member now shows the **Owner** badge; your row shows **Member**. The owner-only controls (invite/transfer/dissolve) are no longer available to you, and a **Leave** button now is.

QA-HH-06 — Owner with other members cannot directly leave/dissolve ● Edge (Web)

Walkthrough: as owner **with** other members present, confirm the bottom card shows the **Transfer ownership** control (with "you must transfer before leaving" note) — **not** a leave/dissolve button. The single-owner invariant is enforced in the UI.

QA-HH-07 — Sole owner dissolves the household ● Core (Web)

Gherkin

```
Given I am the only member and owner of my household
When I click "Leave and delete household" and confirm
Then the household is dissolved and I am re-homed to a fresh solo household
```

Walkthrough

- 1 As sole owner (no other members), bottom card → **Leave & delete household**.
- 2 **Expected:** a confirm dialog warning the household will be deleted; on confirm, the app reloads and you land signed in with a **new empty household you own** (you're never left tenant-less).

QA-HH-08 — Member leaves the household ● Core (Web)

Gherkin

```
Given I am a non-owner member
When I click Leave and confirm
Then I leave the household and am re-homed to a fresh solo household I own
```

Walkthrough

- 1 As a member, **Household** → **Leave** → confirm.
- 2 **Expected:** app reloads; you now own a brand-new empty household. The household you left still exists for its remaining members (verify as the owner: the leaver is gone from the member list).

QA-HH-09 — Owner promotes a member to admin ● Core (Web)

Precondition: signed in as the owner with at least one **member** present (use the QA-INV flow). **Gherkin**

```
Given I am the owner and another member exists
When I click "Make admin" next to that member
Then their badge changes to Admin
```

Walkthrough

- 1 On **Household**, find a member's row → click **Make admin**.
- 2 **Expected:** success banner ("Role updated."); the member's badge flips from **Member** to **Admin**, and the button becomes **Make member**. (Verify as that user — see QA-HH-11.)

QA-HH-10 — Owner demotes an admin to member ● Core (Web)

Gherkin

```
Given I am the owner and an admin exists
When I click "Make member" next to that admin
Then their badge changes back to Member
```

Walkthrough

- 1 On **Household**, on an **Admin** row → click **Make member**.
- 2 **Expected:** success banner; the badge returns to **Member** and the button becomes **Make admin**.

QA-HH-11 — Admin sees management controls but not role/ownership controls ● Core (Web)

Precondition: signed in as the admin promoted in QA-HH-09. **Gherkin**

```
Given I am an admin (not the owner)
When I open Household
Then I can rename the household and invite/remove members
But I see no promote/demote (role) controls and no transfer/dissolve – only Leave
```

Walkthrough

- 1 As the admin, open **Household**.
- 2 **Expected:** the **rename** field and the **Invitations** card are available; member rows show a **Remove** button but no **Make admin/Make member** buttons (role changes are owner-only). The bottom card shows **Leave** (no Transfer ownership / Leave & delete). The owner's row shows **no** action buttons.

QA-HH-12 — Member sees no management controls ● Edge (Web)

Walkthrough: signed in as a plain **member**, open **Household**. **Expected:** read-only name, no Invitations card, no per-row action buttons (no Remove/role controls), only a **Leave** button — unchanged from QA-HH-02 (a member is never shown management controls regardless of the admin tier).

QA-HH-13 — Owner exports household data ● Core (Web)

Gherkin

```
Given I am the household owner on /household
When I click "Download household data"
Then I get a link to a JSON export of the household's data
```

Walkthrough

- 1 As owner, open **Household** → the **Data** card → **Download household data**.
- 2 **Expected:** a success row with a **Download** link; following it downloads a JSON bundle containing the tenant, members and invitations (and each feature's data). Members/admins don't see the Data card (owner-only). No secrets (invitation token hashes) appear in the file.

QA-HH-14 — Seat quota blocks inviting past the plan limit ● Core (Web)

Precondition: the template ships example seat caps (Free = 3 seats, counting members + pending invites; PlanCatalog). A Free household at its cap (e.g. 3 members, or 2 members + 1 pending invite). **Gherkin**

```
Given my Free plan's seats are all used (members + pending invites)
When I invite another member
Then I'm told my seat limit is reached and to upgrade (nothing is invited)
And raising the limit (Pro plan / editing PlanCatalog) lets the invite through
```

Walkthrough

- 1 As owner of a Free household at the seat cap, **Household** → invite a new email.
- 2 **Expected:** an error — "Your plan's seat limit is reached. Upgrade your plan to invite more members." (HTTP 402); no invitation is created and no email is sent.
- 1 Free up a seat (revoke a pending invite / remove a member) **or** move to a higher-seat plan — the next invite succeeds. (Seats count members **plus** pending invites, so invites can't over-provision.)
- 1 **Note:** limits are PlanCatalog data; null/absent = unlimited. Metered-usage caps (IQuotaService.TryConsumeAsync, monthly) are wired the same way where an app calls them.

8. Web — Invitations & joining ● Core

QA-INV-01 — Owner invites by email; token revealed + email sent ● Smoke (Web)

Gherkin

```
Given I am the household owner
When I invite an email address
Then an invitation appears in the pending list with an expiry date
And a one-time join token is revealed in the UI
And an invitation email with a /join link is delivered to Mailpit
```

Walkthrough

- 1 **Household** → Invitations → enter qa.member@gmail.com → **Invite**.
- 2 **Expected:** a green panel reveals the raw token + a **Copy** button and notes a join link was emailed; the address shows under **Pending** with an expiry date.
- 1 Check Mailpit: an invitation email addressed to that user, containing a /join?token=... link.

QA-INV-02 — Invitee accepts and joins (two-user flow) ● Smoke (Web)

Gherkin

```
Given an invitation exists for a second user
When that user opens the /join link, signs in, and accepts
Then they become a member of the inviter's household
```

Walkthrough

- 1 Copy the /join?token=... link from QA-INV-01.

- 2 In an **incognito window**, open the link.
- 3 **Expected:** "You've been invited... sign in to accept." Click **Sign in to accept**; complete any sign-in method as qa.member@gmail.com.
- 1 **Expected:** after sign-in you're returned to the join flow automatically, it processes, and you see "You're in" with a **Go to household** button.
- 1 Click it → **Household** shows you as a **Member** of "QA House".
- 2 Back in the owner window, reload **Household**: the new member appears and the pending invite is gone.

QA-INV-03 — Already-authenticated user accepts directly ● Core (Web)

Walkthrough: while already signed in as a *different* fresh user, open a valid /join?token=.... **Expected:** it accepts immediately (no sign-in step) and shows success. *(Note: a user already in a household who accepts another invite moves to the new household — verify their old membership is replaced, honoring the one-tenant invariant.)*

QA-INV-04 — Join link with missing token ● Edge (Web)

Walkthrough: open /join with no ?token=. **Expected:** "invalid invitation / missing token" state, no crash.

QA-INV-05 — Join with invalid/expired/used token ● Core (Web)

Gherkin

```
Given an invitation token that is invalid, already used, or expired
When an authenticated user opens its /join link
Then they see an error state, not a successful join
```

Walkthrough: while signed in, open /join?token=gabage123 (or reuse a token already accepted in QA-INV-02). **Expected:** the error state with a **Back to household** link; no membership change.

QA-INV-06 — Invite an existing member is rejected ● Core (Web)

Gherkin

```
Given a user is already a member of my household
When I invite their email again
Then I get an "already a member" error and no duplicate invite is created
```

Walkthrough: as owner, invite the email of someone already in the household. **Expected:** a red "already a member" status (HTTP 409); nothing added to pending.

QA-INV-07 — Regenerate a pending invitation ● Core (Web)

Gherkin

```
Given a pending invitation exists
When I click Regenerate
Then a new token is issued (revealed) and the old token no longer works
```

Walkthrough

- 1 On a pending invite, click **Regenerate**.
- 2 **Expected:** a new token is revealed. The **previous** token's /join link now fails (QA-INV-05), the new one works.

QA-INV-08 — Revoke a pending invitation ● Core (Web)

Gherkin

```
Given a pending invitation exists
When I click Revoke
Then it is removed from pending and its token no longer works
```

Walkthrough: click **Revoke** on a pending invite. **Expected:** "invitation revoked" status; it leaves the pending list; opening its old link gives the error state.

QA-INV-09 — Copy token button ● Edge (Web)

Walkthrough: click **Copy** on a revealed token; paste elsewhere. **Expected:** the token is on the clipboard. (If the browser blocks clipboard access, the token is still visible to copy manually — no error shown.)

9. Web — Settings / linked accounts ● Core

QA-SET-01 — View linked providers ● Core (Web)

Gherkin

```
Given I signed up with Google
When I open /settings
Then Google shows as "Connected" and Microsoft shows a "Link" button
```

Walkthrough: sign in with Google, open **Settings**. **Expected:** a row per provider; the one you used shows a **Connected** badge + **Unlink**; the other shows **Link**.

QA-SET-02 — Link a second provider ● Core (Web)

Gherkin

```
Given I am signed in and Microsoft is not linked
When I click Link on Microsoft and complete consent
Then Microsoft becomes Connected on my account (no new account is created)
```

Walkthrough

- 1 **Settings** → **Link** on Microsoft → complete consent with a Microsoft account ****whose email isn't already used by another app user****.
- 1 **Expected:** returned to Settings with a success banner ("Microsoft linked"); Microsoft now shows **Connected**. You can subsequently sign in with either provider into the ***same*** account.

QA-SET-03 — Linking a provider already used by another account is rejected ● Core (Web)

Gherkin

```
Given a provider identity is already linked to a different user
When I try to link it to my account
Then I get an "already in use" error and it is not linked
```

Walkthrough: try to **Link** a Google/Microsoft identity that another app user already owns. **Expected:** Settings shows an "already in use" banner (`?link_error=in_use`); no link made. ***(Expired link-token path shows `?link_error=expired` — exercise if you can stall the flow past the token lifetime.)***

QA-SET-04 — Unlink a provider ● Core (Web)

Gherkin

```
Given two providers are linked to my account
When I click Unlink on one and confirm the dialog
Then it returns to a "Link" state
```

Walkthrough: with two providers connected, click **Unlink** on one. **Expected:** a confirm dialog ("Unlink this sign-in method?"); on **confirm** the row reverts to **Link**. **Cancelling** the dialog leaves it **Connected** — no change. ***(The confirm fails closed: if the browser dialog can't run, the unlink is cancelled, never silently performed. The same guarded confirm protects remove-member / leave / dissolve in §7.)***

QA-SET-05 — Unlinking your only provider never locks you out ● Edge (Web)

Gherkin

```
Given Google is my only linked provider
When I unlink it
Then I can still sign in by email (magic link / OTP)
```

Walkthrough: unlink your sole provider, sign out, and sign back in with **OTP/magic link** using the same email. **Expected:** you reach the **same** account/household. (Email sign-in is always available by design, so provider removal

can't lock you out.)

QA-SET-06 — Settings requires auth ● Edge (Web)

Walkthrough: signed out, navigate to /settings. **Expected:** redirect to /login.

QA-SET-07 — Delete my account ● Core (Web)

Precondition: use a **throwaway** account (this is destructive). Easiest: a member of another owner's household (so no dissolve). **Gherkin**

```
Given I am signed in on /settings
When I use the Danger zone "Delete my account" and confirm
Then my account and personal data are deleted and I'm signed out
```

Walkthrough

- 1 **Settings** → **Danger zone** → **Delete my account** → confirm the dialog.
- 2 **Expected (member):** account deleted; you're signed out and land on /login. Signing in again creates a brand-new account.
- 1 **Owner with other members:** an error tells you to **transfer ownership first** (nothing deleted).
- 2 **Sole owner:** a second confirm warns it also **dissolves the household**; on confirm, the account + household are deleted.

QA-MFA-01 — Enable two-factor (authenticator TOTP) ● Core (Web)

Precondition: signed in; an authenticator app (Google Authenticator, 1Password, Authy, ...) to hand. **Gherkin**

```
Given I am signed in on /settings with two-factor Off
When I enable it, scan the QR (or enter the key) and confirm with a 6-digit code
Then two-factor turns On and I'm shown one-time recovery codes
```

Walkthrough

- 1 **Settings** → **Two-factor authentication** shows an **Off** badge → **Enable two-factor**.
- 2 A **QR code** renders next to a **manual key** (Base32). Scan it (or type the key) into the app.
- 3 Enter the app's current **6-digit code** → **Verify & enable**.
- 4 **Expected:** a success banner, the badge flips to **On**, and a grid of **recovery codes** appears (shown once). **I've saved my codes** returns to the On state.
- 1 **Wrong code:** an inline error ("that code is incorrect or has expired"); nothing changes.

QA-MFA-02 — Two-factor is required at sign-in ● Core (Web)

Precondition: an account with two-factor **On** (QA-MFA-01). **Gherkin**

```
Given my account has two-factor enabled
When I sign in with an email OTP
Then I'm asked for an authenticator code before the session starts
And entering a valid code completes sign-in
```

Walkthrough

- 1 Sign out. On /login, request an **email code**, enter it.
- 2 **Expected:** instead of landing signed-in, a **second prompt** asks for the authenticator code.
- 3 Enter the current 6-digit code → you're signed in (lands on /).
- 4 **Wrong/expired code:** an inline error; you stay on the step-up prompt (no session).
- 5 **Note:** OAuth and magic-link sign-ins enforce the step-up too (QA-MFA-04); native (MAUI) too (QA-MFA-05).

QA-MFA-03 — Use a recovery code, then disable two-factor ● Core (Web)

Gherkin

```
Given my account has two-factor enabled
When I sign in and enter a recovery code at the step-up
Then sign-in completes (that code is now spent)
And I can disable two-factor from Settings with a valid code
```

Walkthrough

- 1 Sign in as in QA-MFA-02; at the step-up, enter one **recovery code** instead of a TOTP → signs in.
- 2 **Settings** → **Two-factor authentication** (On) → **Disable two-factor** → enter a current TOTP (or another recovery code) → **Disable**.
- 1 **Expected:** the badge flips to **Off**; a subsequent sign-in no longer asks for a second step.

QA-MFA-04 — Redirect logins (OAuth / magic link) enforce the step-up ● Core (Web)

Precondition: an account with two-factor **On**, reachable via an OAuth provider and/or magic link. **Gherkin**

```
Given my account has two-factor enabled
When I sign in with Google/Microsoft or a magic link
Then I am redirected to the authenticator step-up before any session is issued
And entering a valid code completes sign-in
```

Walkthrough

- 1 Sign out. Sign in via **Google/Microsoft** (or click a **magic link**).
- 2 **Expected:** instead of landing signed-in, you arrive on `/login` showing the ****authenticator code prompt**** (the URL carries a one-time `?mfa=` challenge). No session exists yet.
- 1 Enter the current 6-digit code (or a recovery code) → you're signed in (`/auth-callback` → `/`).
- 2 **Security check:** confirm you are **not** signed in until the code is accepted — a wrong/expired code keeps you on the prompt with no session. (This closes the gap where redirect logins skipped MFA.)

QA-MFA-05 — Native (MAUI) sign-in enforces the step-up ● Core (Desktop/Android)

Precondition: an account with two-factor **On**; run the desktop/Android shell (see `docs/MOBILE_TESTING.md`). Native has no magic link — use **email OTP** or **OAuth**. **Gherkin**

```
Given my account has two-factor enabled
When I sign in on the native app with an email code or OAuth
Then the app shows the authenticator step-up in-app before completing sign-in
And a valid code (or recovery code) finishes sign-in
```

Walkthrough

- 1 In the native app, sign in with an **email code** (or a provider).
- 2 **Expected:** the app stays on the login screen and shows the **authenticator code prompt** (it does not sign in yet). No session/token is stored.
- 1 Enter the current 6-digit code (or a recovery code) → you're signed in (lands on `/`).
- 2 **Wrong/expired code:** inline error; you remain on the prompt (no session). Tokens arrive in the response body (native transport), same as a normal native login.

QA-NOTIF-01 — Notification bell + list ● Core (Web)

Note: the template has no built-in producer; to see items, a feature must call `INotificationService.NotifyAsync` (seed one in dev, or exercise a downstream feature that notifies). **Gherkin**

```
Given I am signed in
When I open the notification bell in the header
Then I see my notifications newest-first, with unread ones marked
And the bell shows an unread count when I have unread notifications
```

Walkthrough

- 1 In the header, click the **bell**. **Expected:** a dropdown opens; with none, it reads "You're all caught up."
- 1 With unread notifications present: a red **count badge** shows on the bell; unread rows carry a dot +

bold title; each shows a relative time ("just now", "5m", "3h", "2d").

- 1 Click the backdrop (outside the panel) → it closes.

QA-NOTIF-02 — Mark read / mark all read ● Core (Web)

Precondition: at least one unread notification (see QA-NOTIF-01 note). **Gherkin**

```
Given I have unread notifications
When I click one (and, separately, "Mark all read")
Then that one clears its unread mark and the count drops
And "Mark all read" zeroes the count
```

Walkthrough

- 1 Open the bell → click an **unread** row. **Expected:** its dot/bold clears; the count decrements.
- 2 Click **Mark all read**. **Expected:** the count badge disappears; all rows show as read.
- 3 Reload the page → the counts/read state persist (server-side).

QA-NOTIF-03 — Delivery preferences ● Core (Web)

Gherkin

```
Given I am signed in on /settings
When I toggle the In-app or Email notification switches
Then the choice is saved and survives a reload
```

Walkthrough

- 1 **Settings** → **Notifications** card shows two switches (**In-app**, **Email**), both on by default.
- 2 Toggle one off. **Expected:** it saves immediately (optimistic; reverts with an error if it fails).
- 3 Reload → the switch keeps its new state. (Email-off suppresses the email channel on future notifies; in-app-off suppresses the in-app row.)

10. Web — Localization (i18n) ● Core

QA-I18N-01 — Switch language on the login page ● Core (Web)

Gherkin

```
Given I am on /login in English
When I choose Español in the language switcher
Then the page text renders in Spanish
```

Walkthrough: on `/login`, use the language switcher (bottom of the card) → **Español**. **Expected:** titles, button labels, and prompts switch to Spanish; the choice persists on reload.

QA-I18N-02 — Language persists per user across sessions ● Core (Web)

Gherkin

```
Given I am signed in and set my language to Spanish
When I sign out and sign back in (even in a fresh browser)
Then the app loads in Spanish
```

Walkthrough

- 1 Signed in, switch to **Español**. (This saves the locale to your user record via ``PUT /api/auth/locale`` and into the JWT.)
- 1 Sign out; sign back in.
- 2 **Expected:** app comes up in Spanish — the preference followed the `*user*`, not just the browser.

QA-I18N-03 — In-app UI is fully translated (no English leaks) ● Edge (Web)

Walkthrough: in Spanish, walk Home → Household → Settings → invite flow. **Expected:** all visible labels/buttons/validation/status messages are Spanish; flag any English string that leaks.

QA-I18N-04 — Email language matches the requester's UI language ● Core (Web)

Gherkin

```
Given my UI language is Spanish
When I trigger an OTP / magic link / invitation email
Then the email arrives in Spanish
```

Walkthrough

- 1 Set UI to **Español**. Trigger an OTP (and a magic link, and send an invite).
- 2 In Mailpit, open each. **Expected:** subject + body in Spanish. Repeat in English to confirm both.

10b. Web — Admin console (platform staff) ● Core

Precondition: your account's email must be in the staff allowlist — set `Admin__StaffEmails__0` in the repo-root `.env` (see `.env.example`) and restart the API. Non-staff accounts must **not** see any of this.

QA-ADMIN-01 — Staff sees the console; non-staff don't ● Core (Web)

Gherkin

```
Given I am signed in as a platform-staff user
When I look at the header
Then I see an "Admin" link, and /admin lists every tenant with member counts
And a non-staff user sees no Admin link and is refused at /admin
```

Walkthrough

- 1 Signed in as **staff**: an **Admin** button shows in the header → open it (or go to `/admin`).
- 2 **Expected:** the **Tenants** list shows every tenant (name + member count). Click one → detail panel shows members (name/email + role), subscription status, created date, audit-event count.
- 1 Sign in as a **non-staff** user: **no Admin link**; navigating directly to `/admin` shows "You don't have access to the admin console."

QA-ADMIN-02 — View a tenant is audited in that tenant ● Core (Web)

Gherkin

```
Given I am staff viewing a tenant's detail in /admin
When the detail loads
Then an audit event (admin.tenant.viewed) is recorded in that tenant
```

Walkthrough

- 1 As staff, open a tenant's detail in `/admin`.
- 2 **Expected:** an `admin.tenant.viewed` event is written **in that tenant** (visible to that tenant's own audit trail) — the global tenant filter is never loosened; the read enters the target tenant.

QA-ADMIN-03 — Impersonate a user, then stop ● Core (Web)

Gherkin

```
Given I am staff on a tenant's detail
When I "Sign in as" a member and confirm
Then I browse the app as that user with a persistent impersonation banner
And "Stop impersonating" returns me to my own staff identity
```

Walkthrough

- 1 In a tenant detail, click **Sign in as** on a member → confirm the dialog.
- 2 **Expected:** you land on / **as that user** (their name/household in the header); a yellow **impersonation banner** is pinned at the top; the **Admin** link is hidden while impersonating.
- 1 Click **Stop impersonating**. **Expected:** you're back as yourself (staff); the banner is gone.
- 2 The impersonation token is **short-lived (15 min) and non-refreshable** — a full page reload also

returns you to your own identity. Impersonation is **audited** in the target's tenant.

11. Emails (Mailpit) — branding & content ● Core

Delivery is asynchronous (the outbox dispatcher) — emails land in Mailpit a few seconds after the triggering action, and the send request succeeds regardless (see §1.1). Wait briefly before opening Mailpit; a missing email is a delayed/retrying/dead-lettered outbox message, not a request failure.

QA-MAIL-01 — OTP email is branded & correct ● Core

Gherkin

```
Given I request an OTP code
When I open the email in Mailpit
Then it shows the brand logo, brand colours, the 6-digit code, and the tagline
```

Walkthrough: trigger an OTP; open it in Mailpit. **Expected:** the logo image renders (CID-embedded, not a broken image), brand-green styling, a clearly displayed code, footer wordmark + tagline.

QA-MAIL-02 — Magic-link email ● Core

Walkthrough: trigger a magic link; open in Mailpit. **Expected:** branded layout; a working sign-in button/link; sensible subject.

QA-MAIL-03 — Invitation email ● Core

Walkthrough: send an invite; open in Mailpit. **Expected:** branded layout; the recipient's address; a `/join?token=...` link that works (QA-INV-02).

QA-MAIL-04 — Logo renders in a real client ● Edge

Walkthrough (optional, deliverability): forward/inspect one email in a real Gmail/Outlook client. **Expected:** the CID logo still renders. *(Note: from a non-verified domain, real delivery may land in spam — a domain/DKIM concern, not an app bug.)*

12. Desktop — MAUI / Windows ● Core

The desktop client reuses the same UI; only the **auth transport** (loopback browser flow, body token, OS secure storage) differs. Magic link is intentionally **absent** on native.

QA-DSK-01 — OTP sign-in ● Smoke (Desktop)

Gherkin

```
Given the desktop app is on the login screen
When I request an OTP and enter the code from Mailpit
Then I am signed in within the app (no browser)
```

Walkthrough

- 1 Launch the desktop app → login screen. **Expected:** Google/Microsoft buttons + email field with **Email me a code; no magic-link button** (native hides it).
- 1 Enter an email → request code → get it from Mailpit → enter it.
- 2 **Expected:** signed in, app shell shown, household + name in header.

QA-DSK-02 — Google OAuth via loopback ● Core (Desktop)

Gherkin

```
Given the desktop login screen
When I click Continue with Google and complete consent in the system browser
Then the browser shows a "you can close this" page and the app completes sign-in
```

Walkthrough

- 1 Click **Continue with Google**. **Expected:** your **system browser** opens to Google consent.
- 2 Approve. **Expected:** the browser tab shows a "you can close this tab" page; ****focus returns to the app, now signed in. (Repeat for Microsoft**.)**

QA-DSK-03 — "Remember me" across app restart ● Core (Desktop)

Gherkin

```
Given I am signed in to the desktop app
When I fully close and reopen it
Then I am still signed in
```

Walkthrough: close the app completely; relaunch. **Expected:** lands signed in (refresh token held in Windows secure storage / DPAPI, silently exchanged on startup).

QA-DSK-04 — Household & Settings load (authorized API calls) ● Core (Desktop)

Gherkin

```
Given I am signed in to the desktop app
When I open Household and Settings
Then both load their data without an auth error
```

Walkthrough: open **Household** (members load) and **Settings** (linked providers load). **Expected:** both populate — no spinner-forever, no "couldn't load" error. *(This is the Bearer-token-handler path that previously failed on native; it must work.)*

QA-DSK-05 — Link a provider from Settings ● Edge (Desktop)

Walkthrough: **Settings** → **Link Microsoft** → system browser loopback flow → returns to app. **Expected:** Microsoft shows **Connected**; the app shell was never navigated away.

QA-DSK-06 — Sign out ● Edge (Desktop)

Walkthrough: **Sign out**. **Expected:** back to the login screen; reopening the app does **not** auto-sign-in (secure storage cleared).

QA-DSK-07 — Core household flows work on desktop ● Edge (Desktop)

Walkthrough: spot-check rename, invite (token revealed), and leave on desktop. **Expected:** same behavior as web (§7–8) — the UI is shared.

13. Android — MAUI ● Core

Prereq every run: `**adb reverse tcp:5238 tcp:5238**` + API on the https profile. See docs/MOBILE_TESTING.md.

QA-AND-01 — OTP sign-in ● Smoke (Android)

Gherkin

```
Given the Android app is on the login screen with adb reverse set
When I request an OTP and enter the code from Mailpit
Then I am signed in
```

Walkthrough

- 1 Confirm `adb reverse --list` shows `tcp:5238`. Launch the app.
- 2 Login screen: email field + **Email me a 6-digit code**; Google/Microsoft buttons; ****no magic link****.
 - 1 Enter an email → request code → read it in Mailpit (host) → enter it.
 - 2 **Expected:** signed in, app shell shown.

QA-AND-02 — Google OAuth via custom scheme ● Core (Android)

Gherkin

```
Given the Android login screen
When I tap Continue with Google and complete consent
Then the in-app browser returns to the app via the perezosoftware:// scheme, signed in
```

Walkthrough

- 1 Tap **Continue with Google**. **Expected:** a browser tab opens to Google consent.
- 2 Approve. **Expected:** the tab returns control to the app (perezosoftware://auth intent), now signed in. (Repeat **Microsoft** if its :5238 redirect is registered.)

QA-AND-03 — "Remember me" across app restart ● Core (Android)

Walkthrough: swipe-close the app; reopen. **Expected:** still signed in (refresh token in the Android Keystore). *(If adb reverse was lost on a device reboot, re-run it first — a startup failure after reboot is an environment issue, not an app bug.)*

QA-AND-04 — Household & Settings load ● Core (Android)

Walkthrough: open **Household** and **Settings**. **Expected:** both load their data (the native Bearer path works), same as desktop QA-DSK-04.

QA-AND-05 — Navigation drawer / header is reachable ● Edge (Android)

Gherkin

```
Given I am signed in on Android
When I open the navigation (hamburger)
Then the menu is tappable and not hidden under the status bar
```

Walkthrough: tap the hamburger; use **Household/Settings/Sign out**. **Expected:** the header sits below the status bar (safe-area padding) and every item is tappable.

QA-AND-06 — Core flows on Android ● Edge (Android)

Walkthrough: spot-check language switch, invite (token revealed), and leave. **Expected:** parity with web.

14. Cross-cutting security ● Core

QA-SEC-01 — Tenant isolation ● Smoke

Gherkin

```
Given two households owned by two different users
When user A is signed in
Then A can only ever see A's household, members, and invitations – never B's
```

Walkthrough

- 1 Create household A (owner A) and a separate household B (owner B, different account/incognito).
 - 2 As A, open **Household**. **Expected:** only A's members/invites. Confirm B's data never appears.
- *(Deeper API-level probing — e.g. requesting another tenant's resource id — belongs in automated Api . Tests; this manual case verifies the UI never cross-contaminates.)*

QA-SEC-02 — Protected pages require auth ● Core

Walkthrough: signed out, directly visit /household, /settings. **Expected:** each redirects to /login.

QA-SEC-03 — Session is gone after sign-out ● Core

Gherkin

```
Given I sign out
When I press the browser Back button or reload a protected page
Then I am not able to access it – I am sent to /login
```

Walkthrough: sign out, press **Back** to a protected page / reload it. **Expected:** bounced to /login; no stale authenticated view.

QA-SEC-04 — Native open-redirect guard ● Edge (Desktop/Android)

Context/Expected: the native OAuth flow only honors loopback http callbacks or the configured `perezosoftware://` scheme; arbitrary redirect targets are rejected. This is unit-tested (`NativeRedirectPolicyTests`); no manual action needed unless probing the API directly — record as **covered by automated tests**.

QA-SEC-05 — Server-side hardening (automated) ● Edge

Context/Expected: several invariants are enforced at the API/data layer and verified by `tests/Api.Tests`, not by manual UI steps — record as **covered by automated tests** unless probing the API directly:

- **Tenant write-stamping** — a new tenant-scoped row is stamped with the caller's tenant and a foreign-tenant write is rejected (`TenantStampingInterceptorTests`), so reads *and* writes are tenant-isolated.
- **Refresh-token reuse detection** — replaying an already-rotated refresh token revokes all the user's sessions (`RefreshTokenServiceTests`). Manually observable only by capturing and replaying a refresh cookie/token; out of scope for routine QA.
- **Unverified-email takeover guard** fails closed (`ClaimsExtractorTests / UserServiceTests`).
- **Legacy refresh-cookie self-heal** — a stale `Path=/` refresh cookie left by an older build can shadow the live `Path=/api/auth` cookie and wedge sign-in into a `/refresh 401` → "Authentication Failed" loop (seen in Firefox, due to cookie send-order). Setting the refresh cookie now also emits an expiry for the orphan, so the next successful sign-in / refresh sweeps it automatically — no manual cookie-clearing, and upgraded deployments self-heal (`CookieServiceTests`). *Manual repro needs a planted `Path=/` cookie; record as covered by automated tests. If a tester hits a stuck `/refresh 401` loop in Firefox after a deploy, a single re-sign-in clears it.*

14b. API surfaces — PUBAPI + HOOKS (config-gated; curl / Postman) ● Core

These two surfaces are **off by default** and have **no web UI** by design — they're for machines, so they're QA'd with an HTTP client. Any client works; the steps use `curl`. **Postman:** import `GET /api/public/openapi.json` (once PUBAPI is on) to get a ready collection for the public routes.

Preconditions (do once): in the repo-root `.env` set `PublicApi__Enabled=true` and `Webhooks__Enabled=true`, then restart the API. Management of keys/webhooks is **owner-only**, so sign in as an owner and grab a JWT access token from `POST /api/auth/refresh` (the Swagger "Authorize" button shows one) to call the `/api/apikeys` and `/api/webhooks` management routes below. Base URL in these steps is `https://localhost:7160` (use `-k` for the dev cert).

QA-API-01 — Config gate: surfaces are 404 when disabled ● Edge (curl)

Gherkin

```
Given PublicApi:Enabled and Webhooks:Enabled are false (the default)
When I call any /api/public, /api/apikeys or /api/webhooks route
Then it does not exist (404) — the routes aren't mapped and the API-key scheme isn't added
```

Walkthrough

- 1 With both flags **unset/false**, restart the API and call `curl -k https://localhost:7160/api/public/openapi.json` and `.../api/apikeys` (with a JWT).
- 1 **Expected:** **404** for both. Now set the two flags true, restart, and re-check — they become live (401/200). Leave them **on** for the rest of this section.

QA-API-02 — Mint an API key (shown once) ● Core (curl)

Gherkin

Given I am the owner with PUBAPI enabled
When I create an API key
Then I receive the raw pk_... key exactly once, and listing later shows only its prefix/metadata

Walkthrough

- 1 `curl -k -X POST https://localhost:7160/api/apikeys -H "Authorization: Bearer <JWT>" -H "Content-Type: application/json" -d '{"name":"qa","scopes":["read"]}'`
- 2 **Expected: 201** with a key field like pk_... — **copy it now** (never shown again). GET /api/apikeys lists it with prefix/scopes but **no** key.
- 1 **Non-owner:** repeat as a member/admin → **403**.

QA-API-03 — Call the public API with the key; scopes + tenant-scoping ● Core (curl / Postman)

Gherkin

Given a read-only API key for my tenant
When I call the public API with it
Then whoami returns my tenant, and a write-scoped route is refused (403 insufficient_scope)

Walkthrough

- 1 `curl -k https://localhost:7160/api/public/whoami -H "X-API-Key: pk_..."` → **200**, body shows my tenant_id, the key name, and ["read"].
- 1 `curl -k -X POST https://localhost:7160/api/public/echo -H "X-API-Key: pk_..." -d '{"message":"hi"}'` with the **read-only** key → ****403 insufficient_scope****. (A key created with "write" succeeds.)
- 1 **No/blank/garbage key** → **401**. **Revoked key** (DELETE /api/apikeys/{id}) → **401** afterwards.
- 2 **Postman:** import /api/public/openapi.json, set an X-API-Key header on the collection, run whoami.

QA-API-04 — Per-key rate limit ● Core (curl)

Walkthrough

- 1 Fire whoami with one key ~65 times in a minute (for i in \$(seq 1 65); do curl -k -s -o /dev/null -w "%{http_code}\n" https://localhost:7160/api/public/whoami -H "X-API-Key: pk_..."; done).
- 2 **Expected:** the first 60 are **200**, then **429**. A **second** key still returns **200** (budgets are per key, not shared).

QA-API-05 — Register a webhook + send test + verify signature ● Core (curl)

Precondition: a receiver URL that echoes requests — e.g. create one at <https://webhook.site> and copy it. **Gherkin**

Given HOOKS is enabled and I own the tenant
When I register my receiver and send a test event
Then my endpoint receives a signed POST I can verify with the secret

Walkthrough

- 1 POST /api/webhooks (JWT) with {"url":"<webhook.site URL>","event_types":["ping"]} → **201** with a secret (whsec_...) **shown once** — copy it.
- 1 POST /api/webhooks/{id}/test (JWT) → **200** { "delivered": true, "status_code": 200 }.
- 2 On webhook.site, confirm the request has headers ****X-Webhook-Id, X-Webhook-Event: ping**, and **X-Webhook-Signature: sha256=... Verify: HMAC-SHA256 of the raw body**** with your whsec_... secret equals the signature (any HMAC tool, or the app's WebhookSignature.Compute).
- 1 **Non-owner** management → **403**. **Bad URL / no event types** on create → **400**.

QA-API-06 — Delivery log + replay ● Core (curl)

Gherkin

```
Given I've sent test deliveries to my subscription
When I view its delivery log and replay one
Then I see per-attempt rows, and replay re-POSTs the same event to my endpoint
```

Walkthrough

- 1 GET /api/webhooks/{id}/deliveries (JWT) → a list of attempts, newest first, each with success, status_code, error, event_id.
- 1 POST /api/webhooks/deliveries/{deliveryId}/replay (JWT) → **202**; webhook.site receives the **same** event again (same X-Webhook-Id, so a real receiver can dedup).
- 1 **Failure path (optional)**: point the subscription at a URL that returns 500, send a test → the log shows success:false and the outbox retries with backoff (watch the API logs), dead-lettering after the cap. Replay of an unknown/other-tenant delivery id → **404**.

15. Traceability matrix (feature → cases → API)

| | Key API endpoints | | |
|---|--|--|--|
| UTH-02, DSK-02, AND-02 | GET /api/auth/login/{provider}, GET /api/auth/callback/{provider}, native login/callback/exchange | | |
| 5, 06, MAIL-02 | POST /api/auth/magic-link/send, GET /api/auth/magic-link/verify | | |
| 06, AUTH-03/04, DSK-01, MAIL-01 | POST /api/auth/otp/send, POST /api/auth/otp/verify | | |
| | POST /api/auth/otp/send, .../magic-link/send, .../otp/verify (429) | | |
| DSK-03/06, AND-03, SEC-03, jacy-cookie self-heal) | POST /api/auth/refresh, POST /api/auth/logout, GET /api/auth/me | | |
| 7/08/09 | (send + verify endpoints; /login query states) | | |
| SMK-01 | (provisioned on first auth) | | |
| | GET /api/household, PUT /api/household | | |
| | DELETE /api/household/members/{id}, POST /api/household/leave, POST /api/household/transfer-ownership | | |
| 7/08, MAIL-03 | POST /api/household/invitations, GET /api/household/invitations, POST .../{id}/regenerate, DELETE .../{id} | | |
| 4/05 | POST /api/household/invitations/accept | | |
| DSK-05 | GET /api/auth/logins, POST /api/auth/link/{provider}, DELETE /api/auth/logins/{provider} | | |
| | PUT /api/auth/locale (+ resx) | | |
| , I18N-04 | (SMTP via Mailpit) | | |
| | (all [Authorize] endpoints; write-stamping + reuse detection are automated) | | |
| | GET /health, GET /health/ready | | |
| ses) | async via the outbox dispatcher (OutboxMessages) | | |

| | Key API endpoints | | |
|--|--|--|--|
| Api.Tests (Billing*/Entitlement* pending) | POST /api/billing/checkout, .../portal, .../webhook | | |
| Seat limit blocks invite → 402 upgrade
Api.Tests (ServiceTests) | seats (members + pending invites vs Plan.SeatLimit) enforced on POST /api/household/invitations → 402 seat_limit_reached; metered usage via IQuotaService.TryConsumeAsync (monthly UsageCounter). Limits in PlanCatalog (null = unlimited). | | |
| Api.Tests (WebhookHandlerTests, SubscriptionLapseSweepJobTests); Stripe test triggers | webhook transition into past_due/canceled → owner notification (in-app bell + outbox email, NOTIFY) once; SubscriptionLapseSweepJob (6h) nudges the owner once when a paid period lapses without a webhook (LapseNotifiedAt). Verify with stripe trigger invoice.payment_failed (test mode) → owner sees a billing notification in the bell. | | |
| Api.Tests (TenantDissolveTests) | on tenant dissolve, BillingDataContributor wipes the Subscription projection and enqueues a "billing.cancel" outbox message → IBillingProvider.CancelSubscriptionAsync (a deleted tenant stops being billed). HasDataAsync=false (billing never blocks leaving); export gains a billing section (plan/status/period, no Stripe ids). Manual (Stripe test mode): subscribe a throwaway tenant, delete the account, confirm the Stripe subscription is canceled. | | |
| PublicAPI (curl/Postman) + Api.Tests (ServiceTests, BillingTests); boot-verified on/off | PublicApi:Enabled toggles it. Owner-only /api/apikeys (create → raw pk_... once, list, revoke; Permission.ManageApiKeys); API-key auth scheme mints a tenant_id-scoped principal; demo /api/public/whoami (read scope) + /api/public/echo (write scope) via .RequireApiScope. PUBAPI-2 : per-key rate limit (60/min, isolated per key → 429) + a leak-free public OpenAPI doc at /api/public/openapi.json (only the public routes). Off (default) ⇒ routes 404 . Manual: PublicApi__Enabled=true, mint a key, curl -H "X-API-Key: pk_..." /api/public/whoami; fetch /api/public/openapi.json. | | |
| Webhooks (curl/webhook.site) + Api.Tests (SubscriptionServiceTests, OutboxDeliveryTests, OutboxDeliveryLogTests) + WebhookSignatureTests; boot-verified on/off | Webhooks:Enabled toggles it. Owner-only /api/webhooks (register → signing secret whsec_... once, list, delete, send test ; Permission.ManageWebhooks). IWebhookPublisher.PublishAsync fans out to matching active subs → one "webhook" outbox message each → signed POST (X-Webhook-Signature), retry/dead-letter via the outbox. HOOKS-2 : a delivery log (GET /api/webhooks/{id}/deliveries — one row per attempt, success/status/error) + replay (POST /api/webhooks/deliveries/{id}/replay — re-enqueue the exact payload). Off (default) ⇒ routes 404 . Manual: Webhooks__Enabled=true, register a receiver (e.g. a webhook.site URL), hit send test , view deliveries, replay one. | | |
| Api.Tests (AuditLogTests) | append-only IAuditLog + interceptor | | |

| | Key API endpoints | | |
|--|---|---|----------|
| 1/12 (web roster promote/demote ability/limits); Api.Tests (PermissionsTests, OnServiceTests, RoleManagementTests) | PUT /api/household/members/{id}/role (owner-only; admin↔member, owner via transfer only); permission seam gates tenant writes | | |
| Api.Tests (KFileStorageTests, LoadTokenizerTests, RoleTests, StorageMinioTests [real MinIO], UsageRegistrationTests) | IFileStorage (tenant-scoped keys; local disk / S3-compatible — AWS/MinIO/R2/B2, config-gated); local signed GET /api/files/{token} (expiring, single-key, tenant-checked → 404 on any failure); S3 native presigned URLs | | |
| er Household → Data → Api.Tests (ExportTests) | POST /api/household/export (owner-only ExportData → 403 else; JSON bundle via IFileStorage, signed URL; secret-free, tenant-scoped, audited) | | |
| Settings → Danger zone) + (AccountErasureTests) | DELETE /api/auth/me (wipes identity/PII in one tx; owner-with-members → 400, solo owner → 409 without confirm_dissolve; member removed not re-homed; audited; audit trail survives) | | |
| (Settings Confirm/recovery + disable; step-up auth/magic-link, and native logins) (MfaServiceTests, MfaServiceTests, MfaServiceTests) | `GET | POST /api/auth/mfa/enroll | /confirm |
| 03 (header bell: count/mark-read; Settings preference switches) + Api.Tests (NotificationServiceTests, NotificationFanOutTests) | GET /api/notifications (+ ?before=&limit=), /unread-count, POST /{id}/read, /read-all, and `GET | PUT /api/notifications/preferences — **per-user** (scoped to the caller). NotifyAsync` fans out to in-app + email (outbox-backed) per prefs (default both on). | |
| 03 (staff /admin console: tenant impersonate w/ banner + stop) + (StaffServiceTests, RoleTests) | GET /api/admin/me (staff probe, 200 {is_staff} for any caller — drives the nav/gate), GET /api/admin/tenants (+ /{id}) inspection, POST /api/admin/impersonate/{userId} — platform-staff only (config Admin:StaffEmails; non-staff → 403). Detail enters the target tenant (filter never loosened); impersonation returns a short-lived, non-refreshable token with an impersonated_by claim, audited in the target's tenant . | | |

Per-client coverage: Web = full (all suites). Desktop = DSK-01..07 + shared-UI spot checks. Android = AND-01..06 + shared-UI spot checks. Magic link is **web-only** by design.

16. Sign-off sheet

Record one row per executed case. Build = API/web commit SHA (`git rev-parse --short HEAD`).

| Case ID | Client | Result (P/F/Blocked/N-A) | Tester | Build (SHA) | Date | Notes / defect link |
|-----------|--------|--------------------------|--------|-------------|------|---------------------|
| QA-SMK-01 | Web | | | | | |
| QA-SMK-02 | Web | | | | | |

| Case ID | Client | Result (P/F/Blocked/N-A) | Tester | Build (SHA) | Date | Notes / defect link |
|---------|--------|--------------------------|--------|-------------|------|---------------------|
| ... | | | | | | |

Release gate (suggested): all ● Smoke Smoke + all ● Core Core cases Pass on Web; ● Smoke Smoke Pass on Desktop and Android; no open Critical/High defects. ● Edge Edge cases triaged (Pass or accepted-known-issue).

17. Notes for maintainers

- This plan is the manual counterpart to the automated `tests/E2E.Tests` (Playwright/PHPUnit) suite. That suite now covers the OTP auth happy-path and guards (`AuthFlowTests` over `LoginPage`, reading codes from Mailpit via the `Mailpit` REST client) — see `tests/E2E.Tests/README.md` for the run procedure (it documents the same Mailpit SMTP override noted in §1.1). The Gherkin blocks here are written to be lifted directly into new E2E scenarios — keep the two in sync as automation grows.
- When you add an app-specific domain feature on top of this template, add a matching suite here and a row in the traceability matrix (§15) so "entire functionality" stays honest.
- `docs/FEATURES.md` describes the same JWT-based flows at the design level; this plan is their step-by-step verification. Keep the two in sync when behavior changes.
- **Updated 2026-06-22** for the security/quality remediation: OTP lockout is now ****cumulative per email and resend-proof (QA-AUTH-04); passwordless send/verify are rate-limited (QA-AUTH-11); the Settings Unlink**** now requires a fail-closed confirmation (QA-SET-04); and server-side hardening (write-stamping, refresh-reuse detection, fail-closed takeover guard) is captured as automated coverage (QA-SEC-05).
- **Updated 2026-06-23:** QA-AUTH-02 now documents the **tenant-gated same-email auto-link** rule (Microsoft trusted only on the `consumer's` tenant; work/school + unknown providers fail closed — audit MITI-3); and QA-SEC-05 adds the **legacy refresh-cookie self-heal** as automated coverage (`CookieServiceTests`).
- **Updated 2026-06-26** for the platform-foundation work (JOBS/BILLING/OBS, ADR-006/007/008):
- **Transactional email is now delivered asynchronously** via the outbox dispatcher — a few-second delay, and the send request always succeeds (reliability moved to the background). Affects every email-based case; see the §1.1 and §11 notes. The E2E `AuthFlowTests` (which reads OTP codes from Mailpit) now traverses this async path — confirm it waits/polls for the email rather than assuming instant delivery.
- Added **API health/readiness** endpoints + smoke case **QA-SMK-07** (`/health`, `/health/ready`).
- The **billing API** (checkout/portal/webhook), the **append-only audit log**, and **OpenTelemetry** telemetry are API-/operational-level with **no client UI** — covered by `tests/Api.Tests`, E2E pending; manual cases will follow when UI exists (see §2 + §15).
- **Updated 2026-06-30** for RBAC (ADR-009): the tenant role model gains an ****admin**** tier behind a permission seam (owner > admin > member). RBAC-1 added the seam (no behavior change); RBAC-2 added the **owner-only role-change endpoint** (`PUT /api/household/members/{id}/role`, admin↔member; owner is conferred only via transfer) and hardened member-removal so the **owner can't be removed**. This was API-only at first; **RBAC-3 added the web UI** — the Household roster now has owner-only **Make admin / Make member** controls, and the page is admin-aware (admin can rename + invite/remove members, but sees no role/transfer/dissolve controls). New web cases **QA-HH-09..12**; §7 intro updated for the three-tier model. API coverage stays automated (`tests/Api.Tests`).
- **Updated 2026-06-30** for file storage (ADR-010): an `IFileStorage` seam (tenant-scoped keys, local-disk dev default / S3-compatible prod) with a signed, time-limited download endpoint `GET /api/files/{token}`. **API-only** (no UI consumer yet) — covered by `tests/Api.Tests` (`LocalDiskFileStorageTests`, `FileDownloadTokenizerTests`, `FilesControllerTests`); see §2 + §15. Manual cases will follow when a feature (avatars/attachments) wires it to UI. FILES-3 added the **S3-compatible** backend (AWS/MinIO/R2/B2, config-gated by `Storage:S3:Bucket`), tested against a real `MinIO` container (`S3FileStorageMinioTests`); prod config is documented in `.env.example`. **This completes Wave 1** (RBAC + File storage).
- **Updated 2026-07-01** for GDPR data export (ADR-011, Wave 2): owner-only `POST /api/household/export`

assembles the tenant's data (core + each feature's contributor section) into a JSON bundle stored via file storage and returns a **signed download URL**; **secret-free** (no token hashes), tenant-scoped, audited. **API-only** (no UI button yet) — covered by `Api.Tests (TenantExportTests)`; see §2 + §15.

- **Updated 2026-07-01** for GDPR account erasure (ADR-011): `DELETE /api/auth/me` ("delete my account") wipes the caller's identity/PII (`User/UserLogin/RefreshToken/LoginToken`) in one transaction, single-owner-safe (owner-with-members → transfer first; solo owner → confirm to dissolve; member removed, not re-homed), audited (the audit trail survives — actor ids, never PII). **API-only** — covered by `Api.Tests (AccountErasureTests)`. **GDPR epic complete (export + erasure)**.
- **Updated 2026-07-01** for MFA enrollment (ADR-012, Wave 2): authenticator-app **TOTP** (Otp.NET) — `/api/auth/mfa/*` (enroll → provisioning URI/QR; confirm with a code → enable + one-time recovery codes; disable; status). Secret **encrypted at rest** (Data Protection); recovery codes **hashed + single-use**; MFA rows wiped by account erasure. **API-only** (MFA-1) — covered by `Api.Tests (MfaServiceTests)`. **MFA-2 (login step-up)**: an MFA-on login returns a **signed challenge** instead of a session; `POST /api/auth/mfa/verify` completes it with a TOTP/recovery code. Wired into **OTP-verify + native-exchange**; the OAuth/magic-link **redirect** paths route to a client `/mfa` page (needs the MFA UI) and are a flagged **follow-up** — no template-web user can enable MFA via those paths today (no enrollment UI). Covered by `MfaChallengeServiceTests + MfaLoginServiceTests`. **This completes Wave 2** (GDPR + MFA).
- **Updated 2026-07-01** for in-app notifications (ADR-013, Wave 3): a **per-user** notification center — `GET /api/notifications` (paginated), `/unread-count`, `POST /{id}/read`, `/read-all` — scoped to the caller (never cross-user); features produce via `NotifyAsync` (staged in-app row). Notifications are user PII (wiped by account erasure). **API-only** — covered by `Api.Tests (NotificationServiceTests)`. **NOTIFY-2**: per-user **delivery preferences** (`GET|PUT /api/notifications/preferences`, default both on) + `NotifyAsync` **fan-out** — in-app row + email via the outbox-backed `ISender`, gated by prefs. Covered by `NotificationFanOutTests`. Bell-menu UI is an API-first follow-up. **NOTIFY epic + Wave 3's first item complete**.
- **Updated 2026-07-01** for the platform-staff admin surface (ADR-014, Wave 3): `GET /api/admin/tenants (+ /{id})` — **staff-only** (config allowlist `Admin:StaffEmails`, out-of-band; non-staff → 403). Read-only cross-tenant inspection; the per-tenant detail **enters the target tenant** (the global filter is never loosened) and is **audited in that tenant**. **API-only** — covered by `Api.Tests (PlatformStaffServiceTests, AdminControllerTests)`.
- **Updated 2026-07-01** for admin **impersonation** (ADR-014, ADMIN-2): `POST /api/admin/impersonate/{userId}` (staff-only) returns a **short-lived (15-min), non-refreshable** access token carrying the target's identity + an `impersonated_by` claim; **loudly audited in the target's tenant**. Unknown target → 404. Covered by `AdminControllerTests`. **This completes the ADMIN epic — and the planned platform** (nine epics, ADRs 006–014).
- **Updated 2026-07-01** — UI pass (putting a face on the API-first surfaces). **UI-1 (GDPR)**: the owner **Data** → **Download household data** button on **Household (QA-HH-13)** and **Settings** → **Danger zone** → **Delete my account (QA-SET-07)** — wired to `POST /api/household/export` and `DELETE /api/auth/me` (single-owner-safe, with the dissolve second-confirm). EN/ES localized.
- **Updated 2026-07-01** — **UI-2 (MFA)**: a **Two-factor authentication** card in **Settings** (`MfaCard` component) — enroll (`POST /api/auth/mfa/enroll`) renders a **client-side QR** of the `otpauth://` URI (vendored `qrcode-generator`, MIT, in `Shared.Ui/wwwroot/js/`; the secret never leaves the browser) alongside a manual key, confirm (`/confirm`) reveals one-time **recovery codes**, and **disable** (`/disable`) needs a live code. The **Login** page now handles the `OTP-verify mfa_required` response with a **step-up code prompt** → `POST /api/auth/mfa/verify` (TOTP or recovery code) → `/auth-callback`. **QA-MFA-01..03**; EN/ES localized. Native OTP + OAuth/magic-link step-up remain follow-ups (web-first).
- **Updated 2026-07-01** — **UI-3 (Notifications)**: a **header bell** (`NotificationBell` component) — unread-count badge (polled ~60s), a dropdown list (newest-first, unread dot + relative time), click-to-mark-read and **Mark all read** via `GET /api/notifications[/unread-count]`, `POST /{id}/read`, `/read-all`. A **Notifications** card in **Settings** (`NotificationPrefsCard`) with **In-app/Email** switches (optimistic save, reverts on error) via `GET|PUT /api/notifications/preferences`. **QA-NOTIF-01..03**; EN/ES localized. (No built-in producer — items appear once a feature calls `NotifyAsync`.)
- **Updated 2026-07-01** — **UI-4 (Admin console)**: a staff-only `/admin` page (`AdminConsole`) — tenant

list + detail (members/subscription/audit count) and **Sign in as** (impersonation). Staff detection uses a new **non-gating** probe `GET /api/admin/me ({is_staff})` for any authenticated caller — the only API addition in the UI pass; the allowlist stays config-only, actions still 403 for non-staff, so the header shows an **Admin** link only to staff. Impersonation swaps the in-memory session to the short-lived token (`AuthService.BeginImpersonation`), pins an **impersonation banner** in `MainLayout` (reads the `impersonated_by` claim), and **Stop impersonating** restores the staff identity from the refresh cookie; a reload also reverts (the token is non-refreshable). **QA-ADMIN-01..03**; EN/ES localized. **This completes the UI pass (UI-1..4).**

- Updated 2026-07-01 — MFA-3 (redirect step-up, security fix):** OAuth callback and magic-link verify now route through `IMfaLoginService.CompleteOrChallengeAsync` (like the OTP path) instead of issuing a session directly — closing the gap where an MFA-enabled user could sign in via Google/ magic link and **skip the second factor**. When MFA is on, the server redirects to `/login?mfa=<challenge>` (a signed, single-use, 5-min Data-Protection token — no secret), and `Login.razor` reuses the UI-2 step-up prompt → `POST /api/auth/mfa/verify` → `/auth-callback`. Property covered by `MfaLoginServiceTests` (MFA-on → challenge, never a session); **QA-MFA-04**. The dead `IssueRefreshCookieAsync` helper was removed.
- Updated 2026-07-01 — MFA-4 (native step-up):** the MAUI client now handles the `{mfa_required, challenge}` response on the native **OTP** and **OAuth-exchange** paths. `AuthService.VerifyOtpAsync/SignInWithOAuthAsync` now return a `SignInResult` (Success / Failed / MfaRequired+challenge) and a new `VerifyMfaAsync` completes the step-up (tokens in the body, native transport); `Login.razor`'s native branches reuse the same code prompt. Client-only — no API change; build-verified (native is E2E/manual per `MOBILE_TESTING.md`); **QA-MFA-05**. **MFA is now enforced on every sign-in path, web and native — the epic is fully closed, no open gaps.**
- Updated 2026-07-01 — BILLING-5 (quotas):** `IQuotaService` adds plan **seat** limits (members + pending invites vs `Plan.SeatLimit`, enforced on the invite path → ****402 seat_limit_reached, with an upgrade message in the Household invite UI**) and metered usage** (`TryConsumeAsync` against a monthly, self-resetting `UsageCounter`). Limits are `PlanCatalog` data — null/absent = unlimited, so it's inert until set (template ships example caps: Free 3/3, Pro 10/100). New entity + migration `AddUsageCounter`. Covered by `QuotaServiceTests` (10 cases); **QA-HH-14**; EN/ES. Only **BILLING-6** (trial/dunning) and a billing-dissolve contributor remain from the BILLING epic.
- Updated 2026-07-01 — BILLING-6 (trial/dunning):** the owner-facing reaction to the subscription lifecycle. `IBillingNotifier` notifies the tenant **owner** via the notification center (in-app bell + outbox email). The **webhook** notifies once on a transition into `past_due/canceled` (no spam; inbox dedups). A ****SubscriptionLapseSweepJob**** (6h) nudges once when a paid period lapses without a webhook, recording `Subscription.LapseNotifiedAt` (migration `AddSubscriptionLapseNotifiedAt`) — Stripe stays source of truth, no fabricated status. Covered by `BillingWebhookHandlerTests` + `SubscriptionLapseSweepJobTests`; manual via `stripe trigger`. **The BILLING epic is now complete (1–6)**; only optional follow-ups remain (advance trial nudge; billing-dissolve contributor).
- Updated 2026-07-01 — PUBAPI-1 (public API + API keys, config-gated OFF):** a programmatic surface for machines (the user reversed the earlier "no public API" stance). `ApiKey` (hash-only, `pk_...` revealed once; migration `AddApiKey`); a second **API-key auth scheme** mints a `tenant_id`-scoped principal (so tenant isolation applies for free); owner-only `/api/apikeys` management (`Permission.ManageApiKeys`); demo `/api/public/whoami` (read) + `/echo` (write) gated by `.RequireApiScope`. ****PublicApi:Enabled** (default false) — strong gating: off ⇒ the scheme isn't added and the routes 404.** Covered by `ApiKeyServiceTests` (9); boot-verified on (401 without a key) and off (404). **HOOKS** (outbound webhooks) is the companion outbound half — next.
- Updated 2026-07-01 — HOOKS-1 (outbound webhooks, config-gated OFF):** the outbound integration half. `WebhookSubscription` (url + event types + **encrypted** signing secret, revealed once; migration `AddWebhookSubscription`); `IWebhookPublisher.PublishAsync` fans out to matching active subs → one "webhook" **outbox** message each (durable, retried, atomic with the change); `WebhookOutboxHandler` signs (HMAC-SHA256, X-Webhook-Signature) + POSTs, throwing on non-2xx so the outbox retries/dead-letters. Owner-only `/api/webhooks` (`Permission.ManageWebhooks`) with a synchronous **send test**. ****Webhooks:Enabled** (default false) — off ⇒ routes 404.** Covered by `WebhookSignatureTests` (3) + `WebhookSubscriptionServiceTests` + `WebhookDeliveryTests` (11 in `Webhooks/`); boot-verified on (401) and off (404). **This completes the integration story (PUBAPI inbound + HOOKS outbound), both default-off.**
- Updated 2026-07-01 — PUBAPI-2 (hardening):** per-key rate limiting (`RateLimiting.PublicApiPolicy`

partitions on the key id — 60/min, one key can't exhaust another's budget → 429; `RateLimitingTests` proves per-key isolation) and a **leak-free public OpenAPI doc** (`GET /api/public/openapi.json`, anonymous, emits only the `/api/public` routes so the internal v1 surface is never exposed; boot-verified it serves in Production when enabled and 404s when off).

- **Updated 2026-07-01 — HOOKS-2 (delivery log + replay):** a tenant-facing debug trail. `WebhookDelivery` records **one row per delivery attempt** (event, success, status/error, and the sent body; not `ITenantScoped` — written from the tenant-less dispatcher, read side filters by `TenantId`; migration `AddWebhookDelivery`). Owner routes `GET /api/webhooks/{id}/deliveries` + `POST /api/webhooks/deliveries/{id}/replay` (re-enqueue the exact stored payload, same event id). Covered by `WebhookDeliveryLogTests`. **Candidate HOOKS-3 (not built):** a Blazor management UI for webhooks/API keys.
- **Updated 2026-07-01 — BILLING-7 (dissolve cleanup):** closed the one real gap — deleting a billing-enabled tenant left the Stripe subscription active (still charging). `BillingDataContributor` now wipes the `Subscription` projection on dissolve and **cancels the provider subscription** via a `"billing.cancel"` outbox message (`BillingCancelOutboxHandler` → new idempotent `IBillingProvider.CancelSubscriptionAsync`) — out-of-band, not an external call inside the teardown tx. `HasDataAsync=false` so billing never blocks leaving; export gains a secret-free billing section. Covered by `BillingDissolveTests` (6). **Closes the BILLING epic (1–7).**
- **Updated 2026-07-01 — added §14b — manual QA for the API surfaces (PUBAPI + HOOKS):** curl/Postman cases (QA-API-01..06) for the config gate, minting/using API keys (scopes, tenant-scoping, rate limit), and registering webhooks (send-test, signature verification, delivery log, replay). These surfaces have **no web UI by design** (they're for machines), so this is the human-testable complement to the automated tests — the Postman path is one import of `/api/public/openapi.json` away. Referenced from the §15 PUBAPI/HOOKS rows and the §2 "no client UI" note.